

Raul Cano Portolés

+34 682101060 | rcportoles@gmail.com | linkedin.com/in/raul-cano-portoles | github.com/rcportoles
https://raulcanoportoles.netlify.app/

PROFESSIONAL SUMMARY

Versatile Unity Developer with 3 years of experience specializing in **modular architecture**, **multiplayer systems**, and **XR**. Proven track record in developing internal tools and automating build pipelines (CI/CD) to optimize studio workflows. Expert in performance optimization for standalone VR headsets and real-time network synchronization.

EDUCATION

Institut Pedralbes, Pedralbes, Barcelona <i>Higher National Diploma (CFGS) in Multiplatform Application Development</i> <i>(Video Games & Digital Leisure)</i>	<i>Sep 2021 — Jun 2023</i> <i>GPA: 8.1</i>
Institut Eugeni D'Ors, Vilafranca del Penedés <i>Intermediate Vocational Diploma (CFGM) in Microcomputer Systems and Networks</i>	<i>Sep 2019 — Jun 2021</i> <i>GPA: 9</i>

CERTIFICATIONS

Cisco CCNA 1 Certification <i>CISCO</i>	<i>2021</i>
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EXPERIENCE

Unity Systems Developer | The NET VR March 2026 – Present (Remote)

- **Architected real-time data persistence** and authentication systems integrating **Supabase** with Unity for cloud-based state management.
- **Developed a custom Object Streaming system** to optimize RAM and CPU usage, enabling high-density asset loading in complex environments.
- **Built procedural generation tools** to automate level design, significantly streamlining environment layout and asset placement workflows.
- **Engineered asynchronous systems** for real-time external media delivery and high-performance **REST API** consumption.
- **Implemented scalable multi-user frameworks** and real-time state synchronization to support interactive social environments.
- **Optimized asset pipelines** by automating data extraction and preparation using **Python** scripts to reduce manual overhead.
- **Designed decoupled systems** following **Clean Code** principles to ensure the scalability and long-term maintainability of core logic.

Unity & Gameplay Developer | Ping Pong Tech September 2022 – June 2025 (Hybrid)

- Programmed interactive mechanics and gameplay systems for international exhibitions using **modular and reusable C# architectures**.
- **CI/CD Leadership**: Engineered a custom automation pipeline using scripts and GitHub, reducing manual build management time to zero.
- Developed internal Unity tools, including **Object Pooling systems** and content balancing tools for design teams.
- Integrated computer vision (AI cameras) and external hardware sensors into Unity environments.

PROJECTS

Unfound_VR (Horror VR Experience) | Unity, C#, Visual Studio 2022, VR SDK (Oculus), Git

- Optimized high-fidelity environments for **Oculus/Meta Quest (Standalone)** using advanced profiling and draw call reduction.
- Implemented **Finite State Machines (FSM)** for complex NPC behaviors and clean gameplay flow.
- Modified XR Interaction Toolkits to develop realistic weapon mechanics and locomotion systems.

- Target creation; random movement, scoring system, physics, screamers.

Demo Playable Horror Game | Unity, C#, Visual Studio 2022, Git

- Designed a data-driven inventory and interaction system using **ScriptableObjects**.
- Developed a procedural prop-spawning system with boundary constraints and **Object Pooling** to maintain 90+ FPS.

AWARDS & HONORS

Honorary Mention: *Best Student in Multiplatform Application Development (DAM-Vi).*

Finalist: *14th FPemprèn Edition for Unfound_VR*

SKILLS

Gameplay Systems & Software Architecture

- **Core:** Object-Oriented Programming (OOP), Modular Architecture & Design Patterns (SOLID, Singleton, Observer, State).
- **Unity Advanced:** ScriptableObjects-driven architecture, Finite State Machines (FSM), Event Systems & Triggers.
- **Mechanics:** Advanced Interaction Systems, Physics-based Gameplay, Inventory & UI Systems (TextMeshPro).

Optimization, Debugging & Code Quality

- **Profiling:** Performance Optimization for Mobile/Standalone VR, Draw Call Reduction, Memory Management.
- **Code Health:** Advanced Debugging, Profiling & Problem Analysis, Clean Code principles & Best Practices.
- **Memory:** Object Pooling strategies, Garbage Collection (GC) control.

Multiplayer & Systems Integration

- **Networking:** Unity Netcode for GameObjects, Unity Relay, Client-Server architecture.
- **Advanced Tech:** Real-time External Data/Media Integration, Computer Vision & AI Camera Systems.
- **Hardware:** VR/XR Hardware Integration (Meta Quest, SteamVR), Sensor Integration.

Automation, Pipelines & DevOps

- **CI/CD:** Automated Build Pipelines (GitHub Actions/Custom Scripts), Build Delivery Automation.
- **Tools:** Internal Tool Development (Editor Scripting), Custom Inspector Tools.
- **Control:** Version Control with Git (GitFlow), Basic Systems & NAS Administration, Docker (Basic).

Languages & Engines

- **Engines:** Unity (Expert), Unreal Engine (C++/Blueprints).
- **Languages:** C# (Expert), C++, Python, Java, SQL, HTML/CSS/JavaScript.